

Exercise 2.7. Prove the following. For each, m , n and t are integers.

- (a) If $m \mid n$, then $m^2 \mid n^2$. (e) If $m^3 \mid n$ and $n^4 \mid t$, then $m^{12} \mid t$.
(b) If $m \mid n$, then $m \mid (7n^3 + 13n^2 - n)$. (f) If
(c) If $m \mid n$ and $m \mid t$, then $m \mid (n + t)$. (g) If
(d) If $3 \mid 2n$, then $3 \mid n$. (h) If

Solution.

(a) If $m \mid n$, then $m^2 \mid n^2$.

Since $m \mid n$, by definition there is an integer k such that

$$n = mk.$$

Squaring both sides,

$$n^2 = m^2 k^2.$$

Since k^2 is an integer, it follows that

$$\boxed{m^2 \mid n^2}.$$

(b) If $m \mid n$, then $m \mid (7n^3 + 13n^2 - n)$.

Since $m \mid n$, there is an integer k such that

$$n = mk.$$

Then

$$\begin{aligned} 7n^3 + 13n^2 - n &= 7(mk)^3 + 13(mk)^2 - mk \\ &= 7m^3 k^3 + 13m^2 k^2 - mk \\ &= m(7m^2 k^3 + 13mk^2 - k). \end{aligned}$$

The expression in parentheses is an integer. Therefore,

$$\boxed{m \mid (7n^3 + 13n^2 - n)}.$$

(c) If $m \mid n$ and $m \mid t$, then $m \mid (n + t)$.

Since $m \mid n$, there is an integer a such that

$$n = ma.$$

Since $m \mid t$, there is an integer b such that

$$t = mb.$$

Therefore,

$$n + t = ma + mb = m(a + b).$$

Because $a + b$ is an integer,

$$\boxed{m \mid (n + t)}.$$

(d) If $3 \mid 2n$, then $3 \mid n$.

Since $3 \mid 2n$, there is an integer k such that

$$2n = 3k.$$

Multiplying by 2,

$$4n = 6k.$$

Since $4n = 3n + n$, a simpler way is to use the fact that 2 and 3 are relatively prime. Alternatively, from

$$2n = 3k,$$

we have

$$n = \frac{3k}{2}.$$

Because $2n$ is divisible by 3 and $3 \nmid 2$, the factor 3 must divide n . Hence

$$\boxed{3 \mid n}.$$

(e) If $m^3 \mid n$ and $n^4 \mid t$, then $m^{12} \mid t$.

Since $m^3 \mid n$, there is an integer a such that

$$n = m^3 a.$$

Since $n^4 \mid t$, there is an integer b such that

$$t = n^4 b.$$

Substitute $n = m^3 a$:

$$\begin{aligned} t &= (m^3 a)^4 b \\ &= m^{12} a^4 b. \end{aligned}$$

Since $a^4 b$ is an integer,

$$\boxed{m^{12} \mid t}.$$